

Applications of *Beauveria bassiana* (Hypocreales: Cordycipitaceae) to Control Populations of Spotted Lanternfly (Hemiptera: Fulgoridae), in Semi-Natural Landscapes and on Grapevines

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Abstract

Spotted lanternfly, *Lycorma delicatula* (White), is an invasive Asian insect that was initially found in Berks County, Pennsylvania, in 2014. As of early 2020, this pest had been found in five more eastern states and it is expected to continue to expand its geographical range. *Lycorma delicatula* is highly polyphagous but seems to prefer tree-of-heaven, *Ailanthus altissima*. However, grape growers in Pennsylvania have reported significant damage and loss of vines caused by *L. delicatula* adults. In fall 2018, two fungal entomopathogens (*Beauveria bassiana* and *Batkoa major*) drove localized collapses in *L. delicatula* populations in Berks County, Pennsylvania. In 2019, we tested applications of a commercialized mycoinsecticide based on *B. bassiana* strain GHA on *L. delicatula* populations in a public park in southeastern Pennsylvania. A single application of *B. bassiana* reduced fourth instar nymphs by 48% after 14 d. Applications of *B. bassiana* to *L. delicatula* adults in the same park resulted in 43% mortality after 14 d. *Beauveria bassiana* spores remained viable on foliage for 5–7 d after spraying. We also conducted semi-field bioassays with *B. bassiana* GHA (formulated as BoteGHA and Aprehend) and another mycoinsecticide containing *Isaria fumosorosea* Apopka Strain 97 against *L. delicatula* adults feeding on potted grapes. All the mycoinsecticides killed ≥90% of adults after 9 d using direct applications. Aprehend killed 99% of adults after 9 d with exposure to residues on sprayed grapes. These data show that fungal entomopathogens can help to suppress populations of *L. delicatula* in agroecosystems and natural areas.

Key words: entomopathogenic fungi, biological control, invasive insect, planthopper

Spotted lanternfly, *Lycorma delicatula* (White), is an invasive Asian fulgorid that was first discovered in North America in southeastern Pennsylvania in 2014 (Dara et al. 2015). This large planthopper is native to China, Taiwan, and Vietnam (Liu and Mottern 2017). It spread to South Korea in 2004 (Han et al. 2008, Park et al. 2009) and was found in Japan in 2009 (Kim et al. 2013). *Lycorma delicatula* will feed on >73 different plant species in 32 families, including trees, vines, and other woody and herbaceous plants (Liu and Hartlieb 2019). This species is univoltine and the diversity of plant species fed upon is broader for nymphs earlier in the season and narrows as *L. delicatula* age. One of the most preferred hosts, especially prior to oviposition, is the invasive tree-of-heaven, *Ailanthus altissima* (Mill.) Swingle, which contains cytotoxic alkaloids (Dara

et al. 2015). *Lycorma delicatula* feed through the plant epidermis or bark on the phloem tissues and this can cause oozing wounds and can kill branches when populations are dense (Liu 2019). In later instars, populations causing damage are often aggregated and very abundant (Urban 2020). In addition, the copious amounts of honeydew produced by *L. delicatula* feeding can attract vespids wasps and ants, and also leads to the growth of sooty mold (Liu and Mottern 2017). *Lycorma delicatula* has been reported causing economic damage in vineyards in South Korea and, similarly, in southeastern Pennsylvania, where it has caused economic damage by killing grapevines or reducing yields in vineyards (Leach et al. 2019).

From being present in a single county in Pennsylvania in 2014, *L. delicatula* has spread quickly so that by early 2020 it had been

found in 10 eastern states (80 counties total), and populations were considered established and were quarantined in six of these states (Delaware, Maryland, New Jersey, Pennsylvania, Virginia, and West Virginia) (New York State IPM 2020). While adult *L. delicatula* can fly, it is assumed that much of this dispersal is due to incidental movement of eggs, nymphs, and adults by humans (Urban et al. 2020).

There have been reports of resident natural enemies attacking *L. delicatula*, including the egg-parasitic wasp *Ooencyrtus kuvanae* (Howard) (Hymenoptera: Encyrtidae), which was introduced from Asia to control gypsy moth (*Lymantria dispar* (L.)) (Lepidoptera: Erebiidae) (Liu and Mottern 2017, Liu 2019), and generalist arthropod predators, i.e., predatory hemipterans including the native wheel bug (*Arilus cristatus* (L.)) (Hemiptera: Reduviidae) and predatory stink bug (*Apoecilus cynicus* (Say)) (Hemiptera: Pentatomidae) (Barringer and Smyers 2016), Chinese mantis (*Tenodera sinensis* (Saussure)) (Mantodea: Mantidae) (H. Leach, personal communication), spiders (e.g., *Argiope trifasciata* (Forskål)) (Araneae: Araneidae), and vespid wasps (E. H. Clifton, personal observation), but so far there is little evidence to suggest that these natural enemies significantly reduce *L. delicatula* populations. In contrast, Clifton et al. (2019) described epizootics caused by two native entomopathogenic fungi, *Batkoa major* (Thaxt.) Humber and *Beauveria bassiana* (Bals.-Criv.) Vuill., in dense *L. delicatula* populations in Berks County, Pennsylvania, in October 2018. *Lycorma delicatula* adults that had been killed by *B. major* and *B. bassiana* were also found in 3 and 12 additional locations, respectively, throughout Berks County in 2018 (EHC and AEH, unpublished data). In any location where we found *L. delicatula* killed by *B. major*, we also found cadavers of *L. delicatula* killed by *B. bassiana*.

To date, management of the *L. delicatula* populations in North America involves detection, followed by use of various chemical insecticides. Since *L. delicatula* can be found in many diverse habitats on many different hosts, the registered insecticides and application methods differ for homeowners, vineyards, and park managers, as do economic or social thresholds for treatment. Optimization of control methods is in progress (Leach et al. 2019, Urban 2020) and environmentally friendly control approaches are needed. A study by Leach et al. (2019) evaluated insecticides for controlling *L. delicatula* on peaches and grapes, including some products listed by the Organic Materials Review Institute (OMRI). Unfortunately, the OMRI-listed products were not as effective as the conventional insecticides and, thus, organic producers have limited options for managing *L. delicatula* (Leach et al. 2019).

Entomopathogenic fungi are important pathogens infecting many hemipterans (Araújo and Hughes 2016), especially leafhoppers and planthoppers (Soper 1985). Some species of entomopathogenic fungi are mass-produced and are commercially available for use as mycoinsecticides. One of the fungi known to infect *L. delicatula* during 2018 epizootics, *B. bassiana*, is a well-studied entomopathogen (Feng et al. 1994, Chandler 2017) that is commercially available for use on crops and in forests in the U.S. Little is known about the second fungal pathogen, *B. major*, and it has not been developed as a mycoinsecticide.

In the current study, we applied *B. bassiana* to suppress fourth instar and adult populations of *L. delicatula* in the field. We report on the efficacy of these spray trials and persistence of the sprayed fungal spores, as well as the minimal non-target impacts. We also report the efficacy by contact and from residues of three commercial formulations of mycoinsecticides tested in semi-field studies on *L. delicatula* adults feeding on potted grapevines.

Materials and Methods

Norristown Farm Park Trials

Field Site

Replicated large-plot field evaluations of a commercial formulation of *B. bassiana* strain GHA known as BoteGHA ES (Certis USA, L.L.C., Columbia, MD; EPA registration No. 82074-1) were conducted at Norristown Farm Park located at 2500 Upper Farm Road, East Norriton, PA 19403 (Montgomery County, 40°08'43.8"N, 75°20'32.9"W). Four test sites with high populations of early instar *L. delicatula* and an abundance of preferred host plants, including *A. altissima*, black walnut (*Juglans nigra* L.), oriental bittersweet (*Celastrus orbiculatus* Thunb.), and wild grapes (*Vitis* spp.) were selected from the park's 680 acres in late June 2019. An additional condition in selecting test sites was that they would contain host plants in 10 m wide fencerows bounded by non-host corn and soybean fields rather than larger blocks of wooded areas. These criteria attempted to minimize *L. delicatula* populations moving from untreated areas into the plots during the post-spray evaluation period. In addition, this plot width was chosen because the range of our application equipment was limited to only 10 m from the edge of the plots. To reduce population variation between the four replicate sites, we randomly paired each BoteGHA spray plot within the replicate site with a contiguous water-only sprayed control plot of equal size, in a split-plot design. The closest distance between any two replicate sites was 485 meters. All BoteGHA test plots and their adjacent control plots were each 15.2 m long by 9.1 m wide (Supp Fig. S1 [online only]).

The first spray application was made on 2 July 2019, mostly targeting fourth instar *L. delicatula*. On 14 August 2019, the efficacy of BoteGHA was tested on the more mobile adult *L. delicatula*. For this trial, we expanded the existing plots to 30.5 m long by 9.1 m wide (Supp Fig. S2 [online only]). One of the sites sprayed in July was not used in August because the expanded plot size could not fit between an adjacent parking lot and a bridge (near the field plots in the northwest corner on Supp Fig. S1 [online only]). A new site that also had dense populations of *L. delicatula* and could support the expanded plot size for the August spray trial was established near the previous site (northwest corner on Supp Fig. S2 [online only]).

Mycoinsecticide Applications

To dilute the mycoinsecticide for application, we mixed 6.9 ml of BoteGHA ES per L of water, creating a suspension with approximately 1.05×10^8 spores ml⁻¹. From the roads adjacent to the plots, using a tractor with a three-point hitch to pull a power take-off (PTO) driven air blast orchard/vineyard sprayer (Model CDP14P100P, Durand Wayland, LaGrange, GA) with a GunJet handgun attachment (Model AA43HC, Spraying System Co., Wheaton, IL), the mycoinsecticide was applied to runoff (300 PSI at 19 liters per min) to a height of about 9.1 m and a depth of 9.1 m. Before spraying the mycoinsecticide, the control plots were sprayed first with water.

Analyzing *B. bassiana* Conidial Deposition and Persistence on Foliage

Leaves were sampled to quantify the number of viable conidia per cm² of leaf area immediately after the BoteGHA spray applications. Leaves were also sampled 1, 6, 9, and 14 d after the 2 July trial, and daily for 7 d after application for the 14 August trial. On each of these sampling dates, three sets of 10 leaves per leaflets were collected from random locations within each treatment plot that had been sprayed with BoteGHA. Leaf samples were collected from control plots on the day of spray application only. Samples were taken

from a variety of tree species within the plot, although most samples were from *A. altissima*. Each set of 10 leaves per leaflets was placed in a new, 4-L-zip-lock freezer bag (Walmart #564265379), labeled, and placed in a cooler on ice until processed.

To remove conidia from the surfaces of leaves, 100 ml of sterile 0.1% Tween 80 surfactant (CAS 9005-65-6, Acros Organics, The Hague, Netherlands) was added to each bag. The bags were vigorously agitated by hand for 30 s after adding 0.1% Tween, and then left to soak for 1 h. After soaking, the bag was shaken again vigorously for 15 s, before removing 1 ml of the resulting suspension using a sterile pipette tip. A dilution series using sterile deionized water, up to $10 E^{-2}$, was made for each sample. The surface of 2 replicate 9 cm Petri plates containing sterile selective agar media was inoculated with 0.1 ml from each dilution (0, -1, and -2) and spread with a sterile spreader. Selective medium was modified from [Korosi et al. \(2019\)](#) comprising: 3.9% powdered potato dextrose agar (Oxoid, Hampshire, United Kingdom), 0.016% Syllit (Ceres International, West Chester, PA), 0.01% chloramphenicol >98% (CAS 56-75-7, Sigma-Aldrich, Darmstadt, Germany) and Penicillin-Streptomycin (10,000 units and 10 mg, respectively; P4333, Sigma-Aldrich, Darmstadt, Germany). Penicillin-Streptomycin was added after autoclaving and when the agar had cooled but was still liquid, to avoid denaturation.

Plates were incubated at 25°C in the dark for 5 d for the fungus to form pure, white *B. bassiana* colonies that were easily distinguished from other microbial growth. Count data from the two replicate plates with the most appropriate dilution were logged and used to calculate the number of viable *B. bassiana* conidia cm^{-2} of leaf area.

The total leaf area for each set of 10 leaves was calculated using the phone app 'Leafscan' (<http://www.leafscanapp.com/>). Each leaf was removed from the zip-lock bag and placed on a white sheet of card within a 15 cm square demarked by four corner dots. The app uses the phone camera to photograph and calculate the area of the leaf. Each measurement was logged in the data manager within the app, and the cumulative leaf area from all 10 leaves from each sample bag was calculated.

Impact of Field Application on Densities of Fourth Instar *Lycorma delicatula*

For the fourth instar trial, we used timed visual counts to estimate the total number of *L. delicatula* on trees and other plants in five randomly chosen locations within each plot before spraying, and 12 and 14 d after spraying. Each timed count lasted for 2 min. Two counters estimated *L. delicatula* numbers in each plot on 1 July (1 d before spray) and again on 16 July (14 d after spray). Another counter estimated *L. delicatula* numbers in the plots on 14 July (12 d after spray) using the same method.

Impact of Field Application on *Lycorma delicatula* Adults and Non-targets

Rearing to Assess Adult Mortality

To assess the efficacy of BoteGHA applications to suppress *L. delicatula* adults, we randomly sampled adults from field plots after the August spray trial and reared them on potted *A. altissima* plants for daily monitoring. *Ailanthus altissima* plants were started from seeds and grown in a greenhouse (22.5°C, 16:10 (L:D) h) at Cornell University in 3.8-liter pots with soilless potting media (LM-111, Lambert, Rivière-Ouelle, Québec, Canada) with regular watering and fertilization. Before they were used in experiments, *A. altissima* plants were approximately 0.6–1.0 m tall with a minimum of six leaves on the main stem. On 9 August, the potted *A. altissima* plants were moved to a shade house at the Pfeiffer Turf and Ornamental

Research Farm on the Penn State Berks campus in Reading, Pennsylvania, which is located within the *L. delicatula* quarantine. Plants were watered as needed daily in the shade house.

On 15 August 2019, 1 d after plots were sprayed, live *L. delicatula* adults were randomly sampled from the plots at Norristown Farm Park. Approximately 50 *L. delicatula* adults were collected from each plot and taken to the shade house (approximately 71 km away) on the same day. The 50 *L. delicatula* adults from each plot were divided among five *A. altissima* plants, resulting in approximately 10 adults per plant. Bamboo poles (approximately 1.5 m tall) were used as stakes alongside plant stems to support plastic woven mesh nylon insect rearing bags (L 100 cm × W 66 cm) (BugDorm DC3170, MegaView Science Co., Ltd., Taiwan) enclosing the foliage. The nets were tied off at the bottom of the plant and the base of the bamboo pole to prevent *L. delicatula* escape.

The caged *A. altissima* plants and *L. delicatula* adults were checked daily for mortality from 16 to 25 August and on 29 August. During mortality checks, *L. delicatula* cadavers were removed from cages, transferred to sealed individual 30-ml cups containing a piece of moistened paper towel. Cadavers were checked 7–14 d later for fungal outgrowth by *B. bassiana*. Corrected mortality after 14 d was determined using Abbott's correction ([Abbott 1925](#)). Two plants assigned for the control treatment were not included in the data analysis because they were noticeably smaller (<0.60 m tall), their leaves wilted just a few days after the experiment began, as they might have been damaged in transit to Pennsylvania.

Sampling Dead Adults and Non-targets From Groundcloths

On the same day of spraying plots (14 August), we set up four groundcloths in each plot around *A. altissima* or *J. nigra* trees that hosted populations of *L. delicatula* adults. These groundcloths were used for catching *L. delicatula* and non-targets that fell from trees. Groundcloths were 3.7 × 3.7 m, cut from woven white polypropylene fabric (WGC 12, DeWitt, Sikeston, MO). An incision was made from one end towards the center so that it could be wrapped around the tree trunk base. Groundcloths were attached to bases of tree trunks with a staple gun and the edges of each groundcloth were secured with multiple stakes.

We collected *L. delicatula* cadavers and any non-target arthropods from groundcloths on 18, 21, 24, and 29 August (= 4, 7, 10, and 15 d after spray application). After samples were brought into the lab, each cadaver was checked to verify whether the body was intact. Some of the bodies of these *L. delicatula* cadavers were partially eaten or hollowed out by scavengers, e.g., ants, yellowjackets, and slugs. These partially eaten cadavers and any molted exuviae were excluded from data analysis. To promote fungal outgrowth with large numbers of *L. delicatula* cadavers from each sampling date, *L. delicatula* cadavers from each plot were placed into humid containers. For each of these containers, a piece of plastic lighting louver was cut to fit at mid-height inside of a 13.2-liter food storage container (PFSH-6, Winco, Boise, ID). The bottom of the container was filled with water before dampened paper towels were laid on the piece of lighting louver. *Lycorma delicatula* cadavers were arranged on the paper towels with some space between different cadavers. The container's lid was sealed to maintain 100% RH inside. Containers with these cadavers were kept at room temperature (20–22°C) for 7–14 d to allow for fungal outgrowth. Some cadavers already had *B. bassiana* outgrowth when they were collected from groundcloths. For each plot and sampling date, we recorded the total numbers of *L. delicatula* cadavers, with or without *B. bassiana* outgrowth, and the final percentage of those cadavers that had fungal outgrowth from *B. bassiana* infections.

Non-target arthropods collected on groundcloths were stored at 4°C until they were identified to species, when possible, or to family. After identification, the non-targets were put in 30-ml cups containing a piece of moistened paper towel and sealed with a lid for 7–14 d to allow for fungal outgrowth.

On 4 September, the four trees in each plot with attached groundcloths were treated with injections of dinotefuran to kill surviving *L. delicatula* adults and in order to estimate the density of the remaining *L. delicatula* populations in the plots. We injected 0.79 ml Dinocide (12% dinotefuran, J. J. Mauget Company, Arcadia, CA), per cm of trunk diameter at breast height (DBH). *Lycorma delicatula* feeding from trees injected with dinotefuran would die within 48 h (Urban 2020, B. Walsh, personal communication). *Lycorma delicatula* cadavers were collected from the groundcloths on 5–6 September and stored at 4°C before counting. The counts were adjusted for tree size by dividing the total numbers by tree DBH.

Verifying Mortality From BoteGHA or Native Beauveria With Molecular Tools

For any *L. delicatula* cadaver that was recovered from groundcloths in control plots and showed signs of *Beauveria* fungal outgrowth, before or after it was brought into the lab, we attempted cultures on selective media before DNA extractions and polymerase chain reaction (PCR). Isolates that were hypothesized as *Beauveria* spp. were cultured from *L. delicatula* or non-target cadavers by transferring the fungus to selective agar medium, following Goettel and Inglis (1997). A small surface area containing conidia on the insect cuticle was swabbed with a sterile cotton-tip applicator and conidia that had been collected were transferred to 6-cm Petri dishes containing selective media. The medium, selective for *Beauveria* cultures, was adapted from Chase et al. (1986), with 30 g wheat germ liter⁻¹, autoclaved and then filtered through cheesecloth, before adding 0.25 g liter⁻¹ chloramphenicol, 0.20 g dodine liter⁻¹, 0.01 g crystal violet liter⁻¹, and 15 g agar liter⁻¹ before autoclaving again. Selective media plates were sealed with Parafilm and placed in an incubator (20°C, 0:24 (L:D) h). After 14 d, cultures of *Beauveria* isolates were stored at 4°C until DNA extractions were completed. For positive controls, to verify infection by *B. bassiana* GHA, we diluted a suspension of BoteGHA ES with deionized water and spread it on plates with the same selective media to obtain cultures and extract DNA. We were able to culture *Beauveria* and extract DNA from 6 of the 32 cadavers with fungal outgrowth that were collected from groundcloths in the control plots during 21–29 August. Because there were 250 *L. delicatula* cadavers from BoteGHA plots with *Beauveria* outgrowth and under the assumption that most of them were killed by *B. bassiana* GHA we only created cultures and extracted DNA on a random subset of 26 *L. delicatula* cadavers from these plots; these cadavers were collected from groundcloths between 21 and 29 August.

For an isolate presumed to be a species of *Beauveria*, a small agar plug (<1 mm²) from the culture plate was transferred to a 6-cm Petri

dish containing ~6 ml of sterile potato dextrose broth (Code No. 7585, Neogen, Lansing, MI), sealed with Parafilm, and then placed in an incubator (20°C, 0:24 (L:D) h). After 72–96 h, a piece of mycelium (approximately 10 mg) was collected with sterile forceps and placed into a 1.5 ml bead beater tube. After bead-beating the sample for 60 s at 48 krpm in a Mini-BeadBeater (BioSpec Products, Inc.), fungal genomic DNA was extracted using a DNeasy Plant Mini Kit (Qiagen).

For any isolate that was presumed to be a *Beauveria* sp., bi-directional nucleotide sequences were determined for the B locus (Bloc) nuclear intergenic region. A region of Bloc was amplified and sequenced using the primer pair B22-deg × B3.3R, following PCR conditions described by Rehner et al. (2011). The SAP/EXO protocol was used to clean up reaction mixtures prior to sequencing. Sequencing was done by Cornell University Institute of Biotechnology on an ABI 3730x1. Chromatograms and sequences were edited/trimmed, assembled, and aligned using Geneious Prime software (2019.0.4; Biomatters Ltd.), resulting in contigs with 959–969 base pairs for the Bloc region. We then used ClustalW multiple alignment in Geneious, including *B. bassiana* GHA as a positive control, to verify whether the non-targets and the *L. delicatula* adults collected from groundcloths were killed by *B. bassiana* GHA (in BoteGHA) or from different *B. bassiana* strains that naturally occurred in the field. Sequence data were checked with the National Center for Biotechnology Information (NCBI) nucleotide database (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>).

Mycoinsecticide Trials for Adult *Lycorma delicatula* as Direct Sprays and Residuals on Potted Grapevines

We evaluated the direct contact efficacy of *B. bassiana* GHA formulated as BoteGHA ES at two label-recommended field application rates (2.4 and 4.8 liters concentrate ha⁻¹) and Aprehend (ConidioTec LLC, Centre Hall, PA), which also uses *B. bassiana* strain GHA in a pre-mixed, oil-based formulation (Table 1). Additionally, we evaluated PFR-97 WDG (Certis USA), which uses the fungus *Isaria fumosorosea* Wize, Apopka Strain 97, at label-recommended rates of 1.12 and 2.24 kg ha⁻¹ (Table 1). Further, we evaluated the residual efficacy of the higher rate of BoteGHA and Aprehend after application to grape plants. Trials were conducted at the Penn State Berks campus in Reading, PA. We used Riesling grapes (*Vitis vinifera* L. var. Chardonnay) that were grafted on SO4 and 3309 rootstocks in March 2019. Grapes were grown in 11-liter pots in greenhouses at the Fruit Research and Extension Center (FREC; Biglerville, PA) with regular watering and fertilization. The potted grapes were brought to the Pfeiffer Turf and Ornamental Research Farm on the Penn State Berks campus in May 2019. Fungicides, including Captan, Dithane (mancozeb), and Pristine (pyraclostrobin and boscalid), were applied to grapes at both FREC and at Penn State Berks

Table 1. Mycoinsecticide product dilutions and application rates applied to containers (11.7 cm diameter × 10.5 cm height) with 10 adult *Lycorma delicatula* per container

Product and label field application rate	Conidia or CFU in undiluted product	Dilution of concentrate	Volume of diluted product applied (ml)	Total conidia or CFU applied per replicate
BoteGHA low rate (2.34 liter ha ⁻¹)	2.1 × 10 ¹⁰ conidia ml ⁻¹	5 ml liter ⁻¹ water	2.5	2.6 × 10 ⁸ conidia
BoteGHA high rate (4.68 liter ha ⁻¹)	2.1 × 10 ¹⁰ conidia ml ⁻¹	10 ml liter ⁻¹ water	2.5	5.2 × 10 ⁸ conidia
PFR-97 low rate (1.12 kg ha ⁻¹)	1 × 10 ⁹ CFU g ⁻¹	2.4 g liter ⁻¹ water	2.5	6 × 10 ⁶ CFU
PFR-97 high rate (2.24 kg ha ⁻¹)	1 × 10 ⁹ CFU g ⁻¹	4.8 g liter ⁻¹ water	2.5	4.8 × 10 ⁷ CFU
Aprehend (Ready to use)	2.2 × 10 ⁹ conidia ml ⁻¹	No dilution	0.2	4.4 × 10 ⁸ conidia

until June 2019 to allow fungicide residues to dissipate and prevent interference with bioassays.

One day prior to mycoinsecticide trials, *L. delicatula* adults were collected from field sites near the Penn State Berks campus and they were kept overnight in the shade house feeding on potted grape plants inside of rearing cages (Cat. No. 1462W, BioQuip, Rancho Dominguez, CA). Both types of assays on potted grapes were begun on 21 August 2019. Contact activity was conducted by directly spraying *L. delicatula* adults in groups of 10, contained in 940 ml Kraft Brown Cardboard Deli/Soup Bar containers (diameter: 11.7 cm, height: 13.5 cm) with the test solutions. Mycoinsecticide applications were done in the early morning, to keep *L. delicatula* adults out of direct sunlight during the process of transporting, counting and using them for mycoinsecticide trials. Aqueous-based formulations were applied using Grainger spray bottles (BoteGHA, PFR-97, and water control), and the ready to use, oil-based product Aprehend was applied using the ULV spray applicator supplied by the manufacturer (SKU 30101–50, ConidioTec LLC, Centre Hall, PA). The total amount of spray mixture used for direct contact sprays was 2.5 ml for BoteGHA, and PFR-97 and water control formulations, and 0.225 ml for Aprehend, per container. The volume and application rates were different according to formulation type and the resulting infective units (CFU or conidia) applied per container also differed (Table 1). Ten *L. delicatula* adults were treated per container and 10 replicate applications were conducted for each treatment ($n = 100$ per treatment). To avoid suffocation due to excess spray solution at the bottom of the cups, the cups were placed upside down with polyester netting covering the bottom so that excess solution would drain out. After allowing cups to dry for 10 min, the exposed *L. delicatula* adults were evaluated for mortality. Live individuals were transferred to unsprayed potted grapes covered with nylon mesh insect rearing sleeves (same model used for adults on potted *A. altissima*). Potted grapes were kept outside in a turf area on groundcloth to prevent grass and weed growth in the testing area. The potted grapes were watered when necessary. Adult mortality was assessed at the transfer from the cups to the potted grapes, and 5 days after treatment (DAT) (26 August) and 9 DAT (30 August).

Additionally, the residual efficacy of the higher rates of BoteGHA ($4.68 \text{ liter ha}^{-1}$) and Aprehend (pre-mix ULV; $2.2 \times 10^9 \text{ CFUs ml}^{-1}$) was tested by spraying the grapes to wet using approximately 25 ml of BoteGHA solution ($5.25 \times 10^9 \text{ conidia plant}^{-1}$) and 2.25–2.5 ml Aprehend ($4.95 \times 10^9 \text{ conidia plant}^{-1}$). Plants dried for 30 min before 10 naive *L. delicatula* adults were added to each plant. Ten replicate plants were used per treatment ($n = 100$ per treatment).

Data Analysis

All data analyses were performed in SAS 9.4 (2019). For conidial deposition and persistence on foliage, we compared *B. bassiana* CFUs per cm^2 foliage for different spray trials (2 July vs 14 August) using a one-way analysis of variance (ANOVA) with Fisher's least-significant-difference test, using the PROC ANOVA statement. We performed separate comparisons of CFUs at different time points (0, 1, and 6 DAT). The same model was used to compare *B. bassiana* CFUs in each replicate site (plot #) on day 0 and 1 DAT in the 2 July trial, after observing some differences in conidial deposition. We used the $\log_{10}$ transformation on the numbers of *B. bassiana* CFUs per cm^2 foliage before data analysis.

For the visual counts in the July field trial, we used paired *t*-tests with the PROC TTEST statement to compare the total numbers of *L. delicatula* nymphs in each control plot with the numbers in the

adjacent treatment plots. We used the natural-log transformation on total counts to normalize residuals, and analyses were conducted for each sampling date.

For comparisons of adult *L. delicatula* survival times on potted *A. altissima* plants during the August trial, a Cox proportional hazards model was performed using the PROC PHREG statement.

For the *L. delicatula* cadavers collected from groundcloths, we used a mixed-model ANOVA with the PROC MIXED statement. Treatment (control vs BoteGHA), sampling date (18, 21, 24, 29 August), and their interaction were fixed effects. We included plot and the interaction of block and treatment as random effects. We analyzed the cumulative numbers of *L. delicatula* cadavers as the dependent variable and applied a natural-log transformation to normalize residuals.

For the *L. delicatula* adults treated with mycoinsecticides and reared on potted grapes, multiple comparisons of numbers of dead adults were done at each time point using a one-way ANOVA with Fisher's least-significant-difference test, using the PROC ANOVA statement.

Results

Norristown Farm Park Trials

Conidial Deposition and Persistence on Leaves

Very few contaminating microbes were present on the plates for all samples from the first trial (2 July). However, increasing levels of contamination were present in plates from the second trial (14 August), likely due to the accumulation of honeydew from *L. delicatula* activity and resulting sooty mold and other fungi. Regardless, it was still easy to distinguish and count *Beauveria* colonies on these plates as no inhibition from competing fungi or microbes was observed. There were no quantifiable *B. bassiana* colony forming units (CFUs) on foliage in the control plots immediately after spraying (day 0) for both trials.

On day 0 in the plots sprayed with BoteGHA, *B. bassiana* CFUs on foliage were 2.5×10^4 and 3.0×10^4 CFUs per cm^2 leaf surface for the July and August trials, respectively (Fig. 1A). The *B. bassiana* CFUs on day 0 were not significantly different between the July and August trials ($F_{1,22} = 1.19$, $P = 0.2877$). On day 0 in the July trial, *B. bassiana* CFUs in replicate site #4 were significantly lower than the CFUs for sites #1 and #3 ($F_{3,8} = 4.42$, $P = 0.0412$), but not significantly different than site #2 (Fig. 1B). However, there were no significant differences between replicate sites in the August trial on day 0 (Fig. 1C) ($F_{3,8} = 0.78$, $P = 0.5386$).

One DAT, *B. bassiana* CFUs were 1.6×10^4 and 2.1×10^4 CFUs per cm^2 leaf surface during the July and August trials, respectively (Fig. 1A). The counts of *B. bassiana* CFUs 1 DAT were not significantly different between the July and August trials ($F_{1,22} = 2.19$, $P = 0.1527$). Furthermore, for 1 DAT there were no significant differences between replicate sites in both the July ($F_{3,8} = 1.24$, $P = 0.3587$) and August trials ($F_{3,8} = 0.64$, $P = 0.6076$) (Fig. 1B and C).

Six DAT, the densities of *B. bassiana* CFUs on foliage were 2.7×10^3 and 4.1×10^3 CFUs per cm^2 leaf surface during the July and August trials, respectively (Fig. 1A). Still, the *B. bassiana* CFUs 6 DAT were not significantly different between the two trials ($F_{1,22} = 2.53$, $P = 0.1262$). These slight differences in CFU persistence could probably be attributed to the fact that there was 8.20 cm of rainfall around the field site between 2 and 8 July (0–6 DAT), but only 0.81 cm of rainfall between 14 and 20 August (Fig. 1B and C) (Supp File S3 [online only]). Regardless of varying weather and

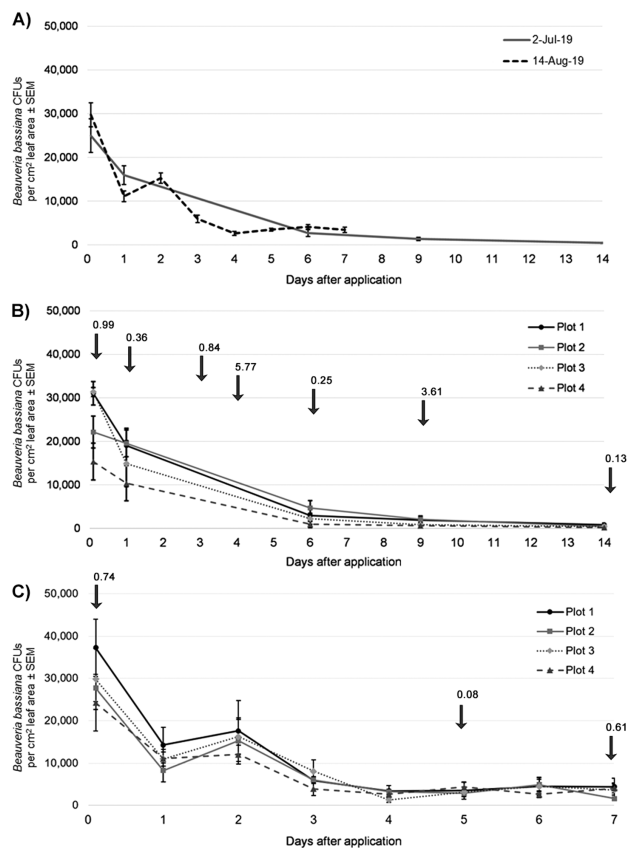


Fig. 1. Persistence of *Beauveria bassiana* colony-forming units on leaves in the field plots at Norristown Farm Park. Error bars represent standard errors of means. (A) Means across replicate sites (plots) comparing the persistence of *B. bassiana* colony-forming units on leaves for the two trials. (B) Leaf samples were taken immediately after application (day 0), and 1, 6, 9, and 14 d after spraying BoteGHA against *Lycorma delicatula* nymphs on 2 July 2019. (C) Leaf samples were taken immediately after application (day 0), and then daily for up to 7 d after application against *L. delicatula* adults on 14 August 2019. Arrows and data labels (aligned above days on x-axis) in (B) and (C) represent total rainfall (cm) for the Philadelphia area (26 km from Norristown Farm Park).

rainfall events, the densities of *B. bassiana* CFUs at 0, 1, and 6 DAT were not significantly different and conidial persistence was similar between trials.

Changes in *Lycorma delicatula* Nymph Population Density

Based on 10 min visual counts of *L. delicatula* nymphs, the average total counts on 1 July (1 d before spray) were 358 ± 95 nymphs in control plots and 327 ± 100 in adjacent BoteGHA plots and these counts were not significantly different (t -test = 3.11, $df = 3$, $P = 0.0530$). For the population densities on 14 July (12 d after spray), the average total 10 min counts were 266 ± 65 in control plots and 142 ± 30 in adjacent BoteGHA plots, and these counts were significantly different (t -test = 6.63, $df = 3$, $P = 0.0070$) (Fig. 2). For population densities on 16 July (14 d after spray), the average total counts were 251 ± 118 in control plots and 130 ± 67 in adjacent BoteGHA plots, and these average counts were significantly different (t -test = 4.82, $df = 3$, $P = 0.0170$) (Fig. 2). Based on these 16 July counts, the density of *L. delicatula* nymphs in BoteGHA plots was 48% lower than those in the adjacent control plots. The average ratio of population densities in control versus BoteGHA plots was 1.13 ± 0.04 on 1 July, but then the ratio increased to 2.14 ± 0.31

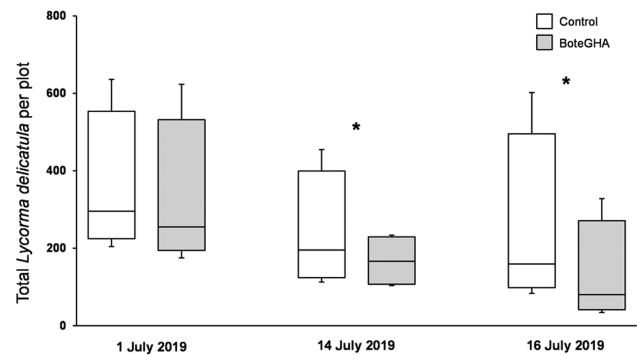


Fig. 2. Visual counts of *Lycorma delicatula* population densities in field plots at Norristown Farm Park. Counts were made 1 d before the 2 July application and again on 14 July (12 d later) and 16 July (14 d later). Box plots show the median (solid line), interquartile range, and range of population densities. Asterisks denote significant differences between treatments based on paired t -tests.

on 16 July. In other words, population densities in the field plots were similar before spraying, but then approximately twice as many *L. delicatula* were found in the control plots compared to adjacent BoteGHA plots 14 d after spraying.

Adult *Lycorma delicatula* Field Trial and Subsequent Bioassays

Survival of Field Collected *Lycorma delicatula* Adults on Potted *A. altissima*

After 14 d on potted plants, 51% of the adults sampled from the BoteGHA plots were dead, compared to 14% mortality for the adults from control plots (Fig. 3). Using Abbott-corrected survival, we estimated that the BoteGHA application resulted in 43% mortality for *L. delicatula* adults after 14 d. The survival times of these adults were significantly different between treatments, based on a Cox proportional hazards model (Log-rank $\chi^2 = 50.63$; $df = 1$; $P < 0.0001$). Among the dead *L. delicatula* adults collected and reared from BoteGHA plots, 83% had fungal outgrowth with *B. bassiana*, whereas only 9% of those from control plots had *B. bassiana* outgrowth.

Dead *Lycorma delicatula* Adults Collected From Groundcloths

The fixed effects of treatments ($F_{1,3} = 30.59$; $P = 0.0116$) and sampling date ($F_{3,18} = 106.47$; $P < 0.0001$) were significant, with significantly more *L. delicatula* cadavers collected from BoteGHA plots over the sampling period compared to control plots (Fig. 4). The interaction of treatment and sampling date was not significant ($F_{3,18} = 1.18$; $P = 0.3435$). By 29 August, we collected a cumulative total of 627 and 1,240 *L. delicatula* cadavers from groundcloths in the control and BoteGHA plots, respectively. Among these dead *L. delicatula* adults, 32 from control plots (5.1% of the total) and 250 from BoteGHA plots (20.2% of the total) had fungal outgrowth from *B. bassiana*. For the final collections of cadavers on 5–6 September, after dinotefuran injections, there was no significant difference between treatments, based on the numbers of adults adjusted for tree size (Supp Fig. S4 [online only]).

Confirmation of *Lycorma delicatula* Mortality From BoteGHA

All six *L. delicatula* cadavers from control plots with successful *Beauveria* cultures and DNA extractions were killed by *B. bassiana* GHA (Supp Table S5 [online only]). We aligned 959 base pairs from the Bloc region and these samples had a 100% match to the positive controls for *B. bassiana* GHA. Twenty-four out of 26 cadavers from

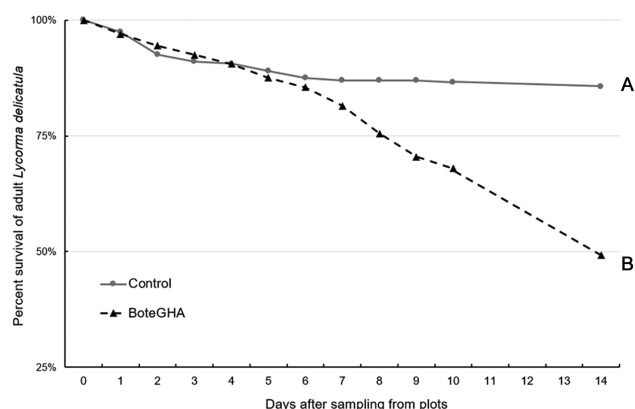


Fig. 3. Survival of adult *Lycorma delicatula* on potted *Ailanthus altissima* plants kept in a shade house. Adults were sampled from field plots in Norristown Farm Park on 15 August (1 d after spraying). Lines represent the fungal applications versus controls. Letters represent significant differences between treatments for survival curves based on Cox proportional hazards.

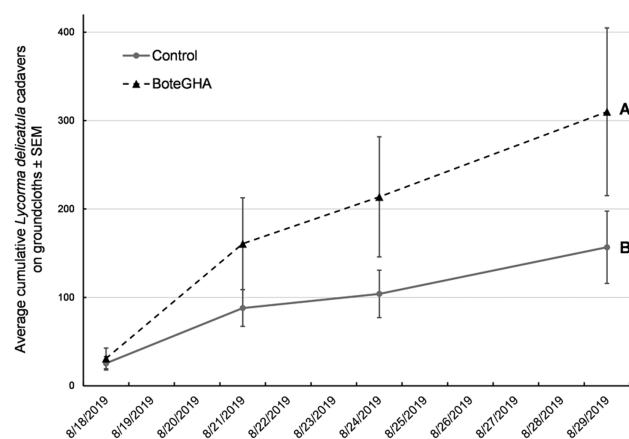


Fig. 4. The mean numbers ($\pm$ standard error) of cumulative *Lycorma delicatula* cadavers recovered from groundcloths in field plots after spray applications on 14 August. The first sampling date was 4 d after application (18 August); the final sampling date was 15 d after application (29 August). Lines represent the fungal applications versus controls. Letters represent significant differences between treatments based on mixed-model analysis of variance.

BoteGHA plots were killed by *B. bassiana* GHA (100% match); however, the other 2 *L. delicatula* adults were killed by native strains of *B. bassiana* (Supp Table S5 [online only]). One of these isolates (*L. delicatula* #50; GenBank MT122124) had a 99.9% match to isolate LHY10A (GenBank JN849669) and four other isolates in the same study, all collected from emerald ash borers, *Agrilus planipennis* Fairmaire (Coleoptera: Buprestidae) (Johny et al. 2012). The other isolate (*L. delicatula* #61; GenBank MT122133) had a 99.8% match to isolate L28A (GenBank JN849673) and six other isolates from the same study, which were also collected from emerald ash borers (Johny et al. 2012). This result confirms that there was some level of naturally occurring infection caused by native strains of *B. bassiana* at Norristown Farm Park in August 2019.

Non-targets From Groundcloths

Among 1,867 *L. delicatula* cadavers that were collected from groundcloths between 18 and 29 August 2019, we found only 59 non-target insects, and only 9 of those were confirmed to be killed by

B. bassiana strain GHA (Supp Table S6 [online only]). One non-target (#46; GenBank MT122109), a tiphiid wasp, was killed by a native *B. bassiana* with 98.2% similarity to isolate CHE-CNRCB 35 (GenBank MH203427) that was isolated from a scarab larva (Serna-Domínguez et al. 2019). Some of the non-targets killed by *B. bassiana* GHA included planthoppers: three individuals were *Flatormenis proxima* (Walker) (Hemiptera: Flatidae) and one was *Acanalonia conica* (Say) (Hemiptera: Acanaloniidae). Some of these native planthoppers have been observed on the same host trees as *L. delicatula* in southeastern Pennsylvania (Fig. 5). Our list of the dead non-target individuals (Supp Table S6 [online only]) collected from field plots includes 22 planthoppers with unknown causes of mortality, 12 of which were *Ormenoides venusta* (Melichar) (Hemiptera: Flatidae). Some of the non-targets were killed by other fungal entomopathogens, including a dog-day cicada (*Neotibicen canicularis* (Harris)) infected by a *Metarhizium* species (Hypocreales: Clavicipitaceae).

Semi-Field Mycoinsecticide Trials With *Lycorma delicatula* Adults on Grapes

All direct contact treatments had significantly greater mortality than controls at 5 DAT ($F_{7,72} = 34.74$, $P < 0.0001$) and 9 DAT (Fig. 6B and C) ($F_{7,72} = 53.91$, $P < 0.0001$). At 9 DAT, direct contact applications killed 90–100% of *L. delicatula* adults and all treatments had significantly greater mortality than controls (Fig. 6C). Direct contact applications of the high rate and low rate of BoteGHA killed 63% and 79% of *L. delicatula* adults 5 DAT (Fig. 6B), and 99 and 100% 9 DAT, respectively (Fig. 6B). The lower rate of BoteGHA had significantly greater efficacy than the higher rate of BoteGHA at 5 DAT (Fig. 6B) and had greater efficacy than the lower rate of PFR-97. PFR-97 killed 58 and 65% of adults 5 DAT (at low and high rates, respectively; Fig. 6B), and 90 and 92% at 9 DAT (at high and low rates, respectively; Fig. 6C). By contrast, 100% of adult *L. delicatula* exposed to Aprehend by direct spray application died within 10 min after exposure (Fig. 6A).

Exposure of *L. delicatula* adults to BoteGHA applied as a residual treatment on grape plants killed 27 and 57% of adults at 5 and 9 DAT, respectively (Fig. 6D and E), whereas adults exposed to plants treated with Aprehend resulted in 80 and 99% mortality at 5 and 9 DAT, respectively (Fig. 6D and E). At both 5 and 9 DAT, residual exposure to Aprehend had significantly higher mortality than BoteGHA.

Discussion

Fourteen days after spraying at Norristown Farm Park, densities of fourth instar *L. delicatula* were 48% lower in BoteGHA-treated plots compared to control plots, based on visual counts. One major challenge in determining the efficacy of *B. bassiana* applications for our study was that researchers are still developing methods for estimation of *L. delicatula* population densities in the variety of landscapes that *L. delicatula* occupies (Urban 2020). Detection lag, population estimates, and development of control strategies are challenges for all invasive species (Mehta et al. 2007). Additionally, movement of *L. delicatula* into the plots may have reduced the apparent impact of BoteGHA in this study because at least some nymphs from the edges of the plots and from the controls probably moved into the treated plots after the residues of BoteGHA diminished, inflating estimates of survival. After the establishment of our trial, recent studies have found that fourth instar nymphs can move >50 m in natural landscapes over the course of 1 wk (K. Hoover, personal communication).

In the August trial, we quantified higher numbers of *L. delicatula* cadavers (= more mortality) in BoteGHA plots compared with controls (Fig. 4). However, by the time the surviving adults that remained on trees were killed using dinotefuran injections on 4 September, there were no significant differences in the numbers of *L. delicatula* adults (Supp Fig. S4 [online only]). Late August to early September precedes a period of increasing aggregation and mating by *L. delicatula* adults in southeastern Pennsylvania (J. Urban and H. Leach, personal communication). One study reported that *L. delicatula* adults can fly ~40 m during one flight bout after launching from trees (Baker et al. 2019). Based on these recent studies and the fact that the dinotefuran injections were made 3 wk after the August application, it is very likely that some of the *L. delicatula* populations moved in and out of our field plots between the dates of spraying and sampling cadavers of dinotefuran-killed *L. delicatula*. Future studies will hopefully benefit from increasing knowledge about instar-specific dispersal of *L. delicatula* and improved methods for quantifying *L. delicatula* densities by instar, especially because *L. delicatula* behavior can vary among different instars (Urban 2020).

Lycorma delicatula adults were sampled from plots and reared on *A. altissima* plants as another method to measure the efficacy of BoteGHA applications during the August spray trial. We had greater confidence in this method because the adults collected just 1 d after spraying would represent the population that had direct and/or 1 d of residual exposure to *B. bassiana* and had limited time to disperse. The experiment with potted *A. altissima* plants showed that 43% of the adults from BoteGHA plots had died by 14 d, after adjusting for the 14% mortality in control plots (Fig. 3). It should be mentioned that there was a heat wave in Reading, PA (where the shade house used for rearing was located) on 18–22 August (3–7 d after sampling; Fig. 3), with high temperatures of 31.5°C or more on five

consecutive days (Supp File S3 [online only]). High temperatures like these can be fungistatic to *B. bassiana*, even after the infective propagules have invaded hosts (Zimmermann 2007). More *L. delicatula* adults died from infections when temperatures cooled down following 22 August (8 d after spraying). The weather records for Philadelphia (25 km from Norristown Farm Park, which was closer than Reading) show that the high temperatures were 32°C or higher on 18–22 August (Supp File S3 [online only]), so the environmental conditions at the application site (Norristown Farm Park) were consistent with the rearing site (Reading).

The observed persistence of *B. bassiana* CFUs on foliage (Fig. 1) corresponds with BoteGHA's label recommendations to reapply when needed every 5 to 7 d. Mycoinsecticides that use fungal entomopathogens like *B. bassiana* perform best when there is direct contact with target pests (Jaronski 2010). The formulation of a mycoinsecticide also plays an important role in its efficacy and residual activity (Jenkins and Thomas 1996, Inglis et al. 2000). Also, environmental conditions such as UV radiation, extreme temperatures, and rainfall can all affect the rate of degradation for *B. bassiana* CFUs on plant surfaces (Inglis et al. 2000, Jaronski 2010).

Based on the molecular data that found some different strains of *B. bassiana* infecting *L. delicatula*, some naturally occurring mortality caused by native *B. bassiana* occurred in Norristown Farm Park. A 2019 survey of *L. delicatula* adults at one field site in Berks County, Pennsylvania (neighboring Montgomery County, the site of Norristown Farm Park) found that as many as 22% of adults died from naturally occurring *B. bassiana* infections during mid-October (E. H. Clifton, unpublished data), but the prevalence of these infections during July–August has not been studied. We used molecular identification tools on a subsample of *B. bassiana* isolates among the 282 *L. delicatula* cadavers that were collected from groundcloths



Fig. 5. A flatid planthopper, *Acanalonia conica*, observed near *Lycorma delicatula* adults on *Ailanthus altissima*. Photo taken on 1 October 2019 in a public park in Reading, Pennsylvania. Photographer: Eric H. Clifton.

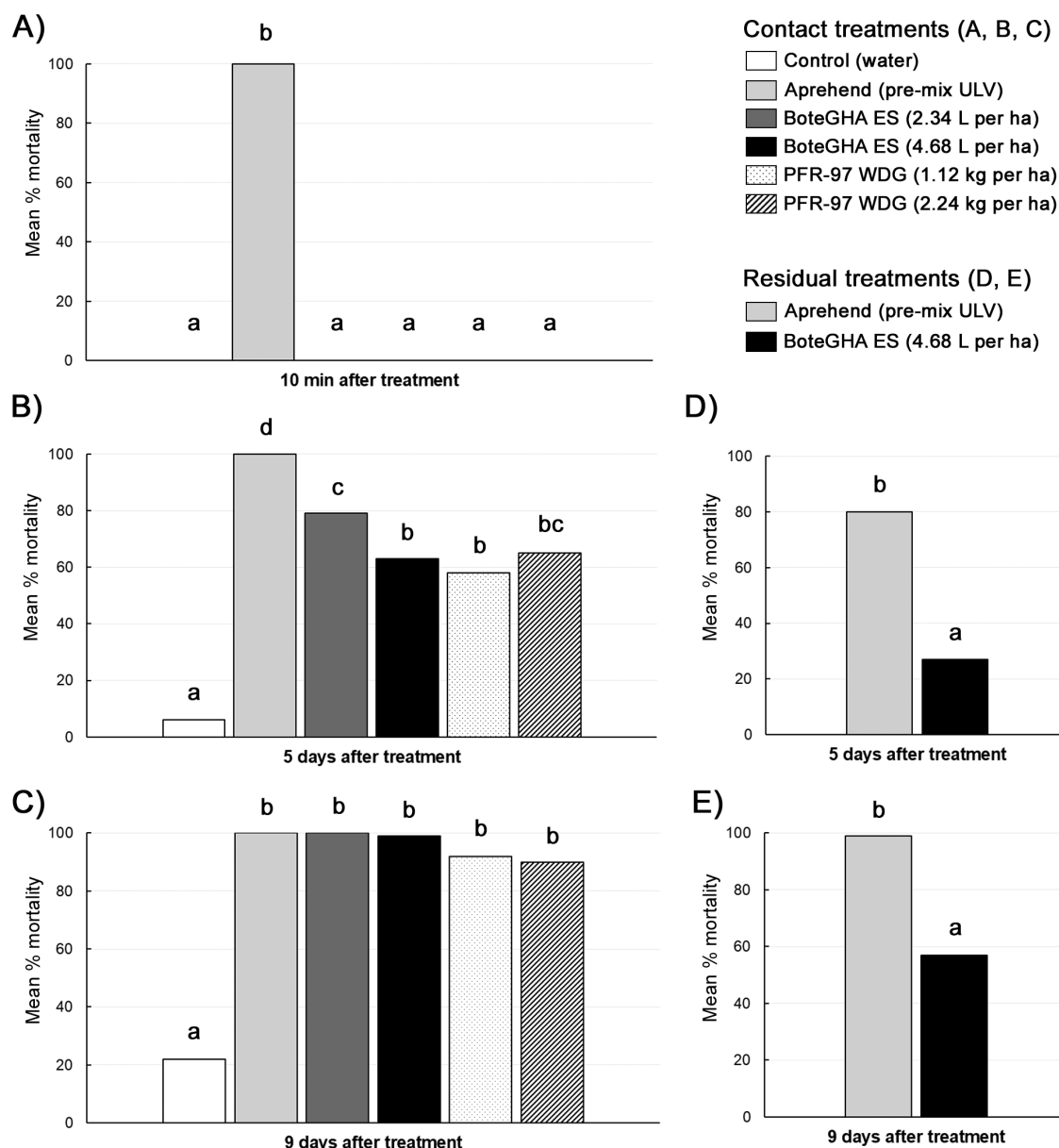


Fig. 6. Mycoinsecticide trials with adult *Lycorma delicatula* on potted grapes. Bar shading indicates treatment. Letters above bars represent significant differences between treatments on a day of sampling, based on multiple comparisons with Fisher's least-significant-difference test. (A) 10 min after treatment (contact). (B) Mean mortality 5 DAT (contact). (C) Mean mortality 9 DAT (contact). (D) Mean mortality 5 DAT (residual). (E) Mean mortality 9 DAT (residual).

and had fungal outgrowth. It should be noted that not every insect killed by *B. bassiana* would show fungal outgrowth, especially for cadavers that laid on groundcloths for multiple days before sampling. Cadavers with multiple days of exposure to scavengers and saprophytes in the environment, UV radiation, and other abiotic factors could fail to show sporulation (Lacey and Solter 2012). This helps to explain why many of the cadavers from treatment plots did not have fungal outgrowth. Our collections from groundcloths probably underestimated the level of mortality caused by applications of BoteGHA.

Two out of 26 *L. delicatula* adults (~8%) from plots that were treated with BoteGHA were killed by native strains of *B. bassiana* during August 2019. It is not known where the treated adults killed by native *B. bassiana* became infected. Although we had a limited sample size for identifying the fungi on cadavers from the control plots, 6 out of 6 (100%) were killed by *B. bassiana* GHA. This

finding may reflect some flaws in the field plot design and our methodology of collecting cadavers from groundcloths over a period of 2 wk. Two weeks would allow time for exposed *L. delicatula* to move from treated plots to adjacent control plots. The maximum size of field plots was somewhat limited because the trials were conducted in a public park with a mixed landscape consisting of pathways, forests, and agricultural production.

We did not observe a great deal of non-target mortality after spraying plots with BoteGHA. Previous studies have shown that *B. bassiana* strain GHA and other strains had minimal impacts on beneficial insects, including parasitoids and generalist predators (Zhu and Kim 2012, Herrick and Cloyd 2017, Martínez-Barrera et al. 2019, Mingotti Dias et al. 2019). Some of the dead insects we found on groundcloths were planthoppers in the families Flatidae and Acanaloniidae (Supp Table S6 [online only]). For many of these non-targets there was no fungal outgrowth. Without fungal

outgrowth from *B. bassiana*, we cannot determine whether BoteGHA killed these insects or if they died from other causes. Searches for ‘*Beauveria*’ and either ‘*Flatormenis proxima*,’ ‘*Ormenoides venusta*,’ or ‘*Acanalonia conica*’ on Web of Science (<https://www.webofknowledge.com>) and Google Scholar (<https://scholar.google.com/>) yielded no results. To our knowledge, there have been no studies exploring the susceptibility of these planthoppers to *Beauveria* spp., but there have been reports of fungal pathogens infecting other planthoppers (Soper 1985). Alternatively, there is little knowledge about the life histories of these native planthoppers and they could have died for other reasons (e.g., the end of the season, as it was late August).

The immediate mortality from direct applications of Aprehend was caused by its oil-based formulation and not from *B. bassiana* infection (Fig. 6A). While it is not likely that large scale field applications would result in direct contact with the entire population, the rapid mortality of those individuals hit directly by spray droplets would be advantageous. Additionally, the enhanced residual efficacy of the Aprehend formulation on plant surfaces after application (Fig. 6D and E) also indicates that Aprehend could play a useful role in *L. delicatula* control operations. Direct applications of BoteGHA and PFR-97 to *L. delicatula* adults did not result in immediate mortality, but rather occurred at later time points after fungal infections progressed (Fig. 6A–C). The PFR-97 product, containing *I. fumosorosea*, showed similar efficacy to BoteGHA, with 90–92% mortality at 9 DAT. Of note is that the concentrations of infective propagules (CFUs) in the PFR-97 direct spray treatments (6×10^6 CFU and 1.2×10^7 CFU) were one or two orders of magnitude lower than the high and low application rates of BoteGHA (2.6×10^8 conidia and 5.2×10^8 conidia) and Aprehend (4.4×10^8 conidia).

Repeated applications of any of these mycoinsecticides that were tested may be necessary to manage large populations of *L. delicatula* adults, especially if they repeatedly enter vineyards from surrounding landscapes. Researchers are still developing economic injury levels (Pedigo et al. 1986) for *L. delicatula* feeding on grapes and other crops. One challenge for grape growers is that *L. delicatula* adults will disperse from the surrounding landscape to enter vineyards, and edge effects have, therefore, been observed (H. Leach, personal communication). There are limited options to stop the adults from repeatedly entering vineyards, but exclusion netting and other tools will be explored in the future. Mycoinsecticides with good residual efficacy, as observed with Aprehend on potted grapes, could be helpful for managing these active *L. delicatula* adults. Targeting applications of fungal entomopathogens to *L. delicatula* adults during periods of aggregation and mating should be investigated further. Besides direct contact with sporulated cadavers, insects exposed to fungal entomopathogens can horizontally transfer infective spores during close contact and mating events (Kreutz et al. 2004, Barbarin et al. 2012, Ugine et al. 2014).

Lycorma delicatula is a particularly difficult pest to manage because of its varying host plant preferences and the many habitats that it can occupy. Previous studies have shown that applications of *B. bassiana* can suppress hemipteran pests on crops and ornamentals (Pogetto et al. 2011, Gouli et al. 2012, Gutierrez-Coarite et al. 2018, Portilla et al. 2019). Other studies have explored the suitability of *Beauveria* spp., including *B. bassiana* strain GHA, for managing forest pests like the emerald ash borer (Liu & Bauer 2008; Castrillo et al. 2010, 2011; Zhang et al. 2011; Mayerhofer et al. 2015). Given the relative efficacy of the PFR-97 and Aprehend treatments in the grapes trial, in spite of the lower CFU concentrations, future trials will also include these products. Interestingly, we have not yet observed *L. delicatula* infected by *Isaria* spp. in the field. Further research is needed to determine the susceptibility of

L. delicatula to different genera and species of fungal entomopathogens. Future studies will consider different formulations and application techniques for *B. bassiana* and *I. fumosorosea*, targeting applications for earlier *L. delicatula* instars, i.e., applications after peak egg hatch, and repeated sprays based on the duration of residual activity. The current study highlights the potential of using *B. bassiana* and *I. fumosorosea* for biological control of high-density *L. delicatula* populations in vineyards and semi-natural landscapes with minimal risks to non-targets.

Supplementary Data

Supplementary data are available at *Environmental Entomology* online.

Acknowledgments

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References Cited

- Abbott, W. S. 1925. A method of computing the effectiveness of an insecticide. *J. Econ. Entomol.* 18: 265–267.
- Araújo, J. P. M., and D. P. Hughes. 2016. Diversity of entomopathogenic fungi: which groups conquered the insect body? *Adv. Gen.* 94: 1–39. doi:10.1016/bs.adgen.2016.01.001.
- Baker, T. C., E. C. Smyers, J. M. Urban, Z. Meng, K. P. Damadaram, A. J. Myrick, M. F. Cooperband, and M. J. Domingue. 2019. Progression of seasonal activities of adults of the spotted lanternfly, *Lycorma delicatula*, during the 2017 season of mass flight dispersal behavior in eastern Pennsylvania. *J. Asia-Pac. Entomol.* 22: 705–713.
- Barbarin, A. M., N. E. Jenkins, E. G. Rajotte, and M. B. Thomas. 2012. A preliminary evaluation of the potential of *Beauveria bassiana* for bed bug control. *J. Invertebr. Pathol.* 111: 82–85.
- Barringer, L. E., and E. C. Smyers. 2016. Predation of the spotted lanternfly, *Lycorma delicatula* (White) (Hemiptera: Fulgoridae) by two native Hemiptera. *Entomol. News.* 126: 71–74.
- Bartlett, C. R. 2018. North American Fulgoridae: lanternflies or fulgorid planthoppers. Available from canr.udel.edu/planthoppers/north-america/north-american-fulgoridae/
- Castrillo, L. A., M. H. Griggs, H. Liu, L. S. Bauer, and J. D. Vandenberg. 2010. Assessing deposition and persistence of *Beauveria bassiana* GHA (Ascomycota: Hypocreales) applied for control of the emerald ash borer, *Agrilus planipennis* (Coleoptera: Buprestidae), in a commercial tree nursery. *Biol. Control.* 54: 61–67.
- Castrillo, L. A., M. H. Griggs, C. M. Ranger, M. E. Reding, and J. D. Vandenberg. 2011. Virulence of commercial strains of *Beauveria bassiana* and *Metarhizium brunneum* (Ascomycota: Hypocreales) against

- adult *Xylosandrus germanus* (Coleoptera: Curculionidae) and impact on brood. *Biol. Control* 58: 121–126.
- Chandler, D. 2017. Basic and applied research on entomopathogenic fungi, pp. 69–89. In L. A. Lacey (ed.), *Microbial control of insect and mite pests*. Academic Press, Amsterdam.
- Chase, A. R., L. S. Osborne, and V. M. Ferguson. 1986. Selective isolation of the entomopathogenic fungi *Beauveria bassiana* and *Metarhizium anisopliae* from an artificial potting medium. *Fla. Entomol.* 69: 285–292.
- Clifton, E. H., L. A. Castrillo, A. Gryganskyi, and A. E. Hajek. 2019. A pair of native fungal pathogens drives decline of a new invasive herbivore. *Proc. Natl. Acad. Sci. U. S. A.* 116: 9178–9180.
- Dara, S. K., L. Barringer, and S. P. Arthurs. 2015. *Lycorma delicatula* (Hemiptera: Fulgoridae): a new invasive pest in the United States. *J. Integr. Pest Mgmt.* 6: 20.
- Feng, M. G., T. J. Poprawski, and G. G. Khachatourians. 1994. Production, formulation and application of the entomopathogenic fungus *Beauveria bassiana* for insect control: current status. *Biocontr. Sci. Technol.* 4: 3–34.
- Goettel, M. S., and G. D. Inglis. 1997. Fungi: hyphomycetes, pp. 213–249. In L. Lacey (ed.), *Manual of techniques in insect pathology*. Academic Press, San Diego, CA.
- Gouli, V., S. Gouli, M. Skinner, G. Hamilton, J. S. Kim, and B. L. Parker. 2012. Virulence of select entomopathogenic fungi to the brown marmorated stink bug, *Halyomorpha halys* (Stål) (Heteroptera: Pentatomidae). *Pest Manag. Sci.* 68: 155–157.
- Gutierrez-Coarite, R., W. P. Heller, M. G. Wright, J. Mollinedo, L. Keith, L. Sugiyama, and S. Chun. 2018. Entomopathogenic fungi as mortality factors of macadamia felted coccid, *Eriococcus ironsidei* (Hemiptera: Eriococcidae) in Hawaii. *Proc. Hawaii Entomol. Soc.* 50: 9–16.
- Han, J.M., H. J. Kim, E. J. Lim, S. H. Lee, Y. J. Kwon, and S. W. Cho. 2008. *Lycorma delicatula* (Hemiptera: Auchenorrhyncha: Fulgoridae: Aphaeninae) finally, but suddenly arrived in Korea. *Entomol. Res.* 38: 281–286.
- Herrick, N. J., and R. A. Cloyd. 2017. Direct and indirect effects of pesticides on the insidious flower bug (Hemiptera: Anthocoridae) under laboratory conditions. *J. Econ. Entomol.* 110: 931–940.
- Inglis, G. D., T. J. Ivie, G. M. Duke, and M. S. Goettel. 2000. Influence of rain and conidial formulation on persistence of *Beauveria bassiana* on potato leaves and Colorado potato beetle larvae. *Biol. Control* 18: 55–64.
- Jaronski, S. T. 2010. Ecological factors in the inundative use of fungal entomopathogens. *BioControl*. 55: 159–185.
- Jenkins, N. E., and M. B. Thomas. 1996. Effect of formulation and application method on the efficacy of aerial and submerged conidia of *Metarhizium flavoviride* for locust and grasshopper control. *Pestic. Sci.* 46: 299–306.
- Johny, S., G. Kyei-Poku, D. Gauthier, K. V. Frankenhuysen, and P. J. Krell. 2012. Characterization and virulence of *Beauveria* spp. recovered from emerald ash borer in southwestern Ontario, Canada. *J. Invertebr. Pathol.* 111: 41–49.
- Kim, H., M. Kim, D. H. Kwon, S. Park, Y. Lee, J. Huang, S. Kai, H. Lee, K. Hong, Y. Jang, and S. Lee. 2013. Molecular comparison of *Lycorma delicatula* (Hemiptera: Fulgoridae) isolates in Korea, China, and Japan. *J. Asia Pac. Entomol.* 16: 503–506.
- Korosi, G. A., B. A. L. Wilson, K. S. Powell, G. J. Ash, A. Reineke, and S. Savocchia. 2019. Occurrence and diversity of entomopathogenic fungi (*Beauveria* spp. and *Metarhizium* spp.) in Australian vineyard soils. *J. Invertebr. Pathol.* 164: 69–77.
- Kreutz, J., G. Zimmermann, and O. Vaupel. 2004. Horizontal transmission of the entomopathogenic fungus *Beauveria bassiana* among the spruce bark beetle, *Ips typographus* (Col., Scolytidae) in the Laboratory and under Field Conditions. *Biocontr. Sci. Technol.* 14: 837–848.
- Lacey, L. A., and L. F. Solter. 2012. Initial handling and diagnosis of diseased invertebrates, pp. 1–12. In L. A. Lacey (ed.), *Manual of techniques in invertebrate pathology*, 2nd ed. Academic Press, London, United Kingdom.
- Leach, H., D. J. Biddinger, G. Krawczyk, E. Smyers, and J. M. Urban. 2019. Evaluation of insecticides for control of the spotted lanternfly, *Lycorma delicatula* (Hemiptera: Fulgoridae), a new pest of fruit in the Northeastern US. *Crop Prot.* 124: 104833. doi:10.1016/j.cropro.2019.05.027.
- Liu, H. 2019. Occurrence, seasonal abundance, and superparasitism of *Ooencyrtus kuvanae* (Hymenoptera: Encyrtidae) as an egg parasitoid of the spotted lanternfly (*Lycorma delicatula*) in North America. *Forests*. 10: 79. doi:10.3390/f10020079.
- Liu, H., and L. S. Bauer. 2008. Microbial control of emerald ash borer, *Agilus planipennis* (Coleoptera: Buprestidae) with *Beauveria bassiana* strain GHA: Greenhouse and field trials. *Biol. Control* 45: 124–132.
- Liu, H., and R. J. Hartlieb. 2019. Spatial distribution of *Lycorma delicatula* (Hemiptera: Fulgoridae) egg masses on tree-of-heaven, black walnut, and siberian elm in North America. *J. Econ. Entomol.* 112: 350. doi:10.1093/jeet/toz350.
- Liu, H., and J. Mottern. 2017. An old remedy for a new problem? Identification of *Ooencyrtus kuvanae* (Hymenoptera: Encyrtidae), an egg parasitoid of *Lycorma delicatula* (Hemiptera: Fulgoridae) in North America. *J. Ins. Sci.* 17: 1–6.
- Martínez-Barrera, O. Y., J. Toledo, P. Liedo, J. Gómez, J. Valle-Mora, J. Cancino, and P. Montoya. 2019. Does *Beauveria bassiana* (Hypocreales: Cordycipitaceae) affect the survival and fecundity of the parasitoid *Coperta haywardi* (Hymenoptera: Diapriidae)? *Environ. Entomol.* 48: 156–162.
- Mayerhofer, J., J. Enkerli, R. Zelger, and H. Strasser. 2015. Biological control of the European cockchafer: persistence of *Beauveria brongniartii* after long-term applications in the Euroregion Tyrol. *BioControl*. 60: 617–629.
- Mehta, S. V., R. G. Haight, F. R. Homans, S. Polasky, and R. C. Venette. 2007. Optimal detection and control strategies for invasive species management. *Ecol. Econ.* 61: 237–245.
- Mingotti Dias, P. E. de Souza Loureiro, L. G. Amorim Pessoa, F. Mendes de Oliveira Neto, R. A. de Souza Tosta, and P. E. Teodoro. 2019. Interactions between fungal-infected *Helicoverpa armigera* and the predator *Chrysoperla externa*. *Insects*. 10: 309.
- New York State IPM (Integrated Pest Management). 2020. Confirmed spotted lanternfly locations. <https://nysipm.cornell.edu/environment/invasive-species-exotic-pests/spotted-lanternfly/>
- Park, J., M. Kim, S. Lee, S. Shin, J. Kim, and I. Park. 2009. Biological characteristics of *Lycorma delicatula* and the control effects of some insecticides. *Korean J. Appl. Entomol.* 48: 53–57.
- Pedigo, L. P., S. H. Hutchins, and L. G. Hingley. 1986. Economic injury levels in theory and practice. *Annu. Rev. Entomol.* 31: 341–368.
- Pogetto, M. F. H. A. D., C. F. Wilcken, M. J. Gimenes, R. S. Christovam, and E. P. Prado. 2011. Control of red-gum lerp psyllid with formulated mycoinsecticides under semi-field conditions. *Int. J. Trop. Insect Sc.* 31: 85–91.
- Portilla, M., H. K. Abbas, C. Accinelli, and R. Luttrell. 2019. Laboratory and field investigations on compatibility of *Beauveria bassiana* (Hypocreales: Clavicipitaceae) spores with a sprayable bioplastic formulation for application in the biocontrol of tarnished plant bug in cotton. *J. Econ. Entomol.* 112: 549–557.
- Rehner, S. A., A. M. Minnis, G. H. Sung, J. J. Luangsa-ard, L. Devotto, and R. A. Humber. 2011. Phylogeny and systematics of the anamorphic, entomopathogenic genus *Beauveria*. *Mycologia*. 103: 1055–1073.
- Serna-Domínguez, M. G., G. Y. Andrade-Michel, R. Rosas-Valdez, P. Castro-Félix, H. C. Arredondo-Bernal, and A. Gallou. 2019. High genetic diversity of the entomopathogenic fungus *Beauveria bassiana* in Colima, Mexico. *J. Invertebr. Pathol.* 163: 67–74.
- Soper, R. S. 1985. Pathogens of leafhoppers and planthoppers, pp. 469–488. In L. R. Nault and J. G. Rodriguez (eds.), *The leafhoppers and planthoppers*. Wiley, New York, NY.
- Ugine, T. A., K. E. Peters, S. Gardescu, and A. E. Hajek. 2014. The effect of time postexposure and sex on the horizontal transmission of *Metarhizium brunneum* conidia between Asian longhorned beetle (Coleoptera: Cerambycidae) mates. *Environ. Entomol.* 43: 1552–1560.
- Urban, J. M. 2020. Perspective: shedding light on spotted lanternfly impacts in the USA. *Pest Manag. Sci.* 76: 10–17.
- Zhang, L. W., Y. J. Liu, J. Yao, B. Wang, B. Huang, Z. Z. Li, M. Z. Fan, and J. H. Sun. 2011. Evaluation of *Beauveria bassiana* (Hyphomycetes) isolates as potential agents for control of *Dendroctonus valens*. *Insect Sci.* 18: 209–216.
- Zhu, H., and J. J. Kim. 2012. Target-oriented dissemination of *Beauveria bassiana* conidia by the predators, *Harmonia axyridis* (Coleoptera: Coccinellidae) and *Chrysoperla carnea* (Neuroptera: Chrysopidae) for biocontrol of *Myzus persicae*. *Biocontr. Sci. Technol.* 22: 393–406.
- Zimmermann, G. 2007. Review on safety of the entomopathogenic fungi *Beauveria bassiana* and *Beauveria brongniartii*. *Biocontr. Sci. Technol.* 17: 553–596.